

ABHISHEK M R

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PROFESSIONAL SUMMARY

Software Engineer specializing in backend architectures, distributed systems, application security platforms, and low-latency C++/Java applications. Upstream contributor to Tier-1 open-source systems (Valkey, Fastify, QuantConnect Lean, QuickFIX). Experienced in building AST data-flow analyzers, deterministic order book matching engines, event-driven microservices with RabbitMQ, and stream backpressure pipelines.

TECHNICAL SKILLS

- Languages:** C++17, Java 21, JavaScript (ES6+), TypeScript, Python 3, C, C#, SQL
- Backend & Distributed:** Spring Boot 3, Node.js, Express, FastAPI, RabbitMQ, WebSockets, RESTful APIs
- Databases & Storage:** PostgreSQL 16/17, MySQL, Redis 7 / Valkey (Pub/Sub & Caching), Prisma ORM
- Systems & Security:** AST Analysis, Source-to-Sink Tracking, SARIF, Limit Order Books, Fixed-Point Math, Stream Backpressure, OCC
- Infrastructure & Tools:** Docker & Compose, Git, GitHub Actions, Linux / POSIX Shell, Vitest, Playwright, Cytoscape.js
- Core CS Foundations:** Data Structures & Algorithms, OOP, OS (Memory, I/O), DBMS, Computer Networks, System Design

UPSTREAM OPEN SOURCE CONTRIBUTIONS

- Valkey — Linux Foundation Key-Value Datastore (Redis Fork) | C, Systems & Storage Internals** [\[valkey-io/valkey\]](https://valkey-io/valkey)
- Resolved stream trimming integer overflow when $\text{MAXLEN} \geq 2^{32}$ on 32-bit builds in `t_stream.c`; removed static compression buffer in RDB serialization to eliminate re-entrancy hazards; refactored key eviction core.
- Fastify Ecosystem (fastify-typebox, fastify-swagger, ajv-compiler) | TypeScript, Web Framework Runtime** [\[github.com/fastify\]](https://github.com/fastify)
- Implemented schema \$ref reference resolution in TypeBox validator compiler; added OpenAPI 3.x path parameter serialization support; updated route validator/serializer compiler TypeScript definitions.
- QuantConnect Lean & QuickFIX (Quantitative & Protocol Engines) | C#, C++, Financial Systems** [\[Lean\]](https://lean.com) [\[QuickFIX\]](https://quickfix.com)
- Fixed non-USD futures cash adjustment in `FutureSettlementModel` and lunch-break market bar calculations in Lean; resolved socket initiator disconnect callback notification on session disconnect in QuickFIX.

SELECTED ENGINEERING PROJECTS

- DontTrust — Application Security & Attack-Surface Intelligence Platform | TypeScript, Node.js, React 19, AST Analysis, Cytoscape.js, SARIF, Docker** [\[GitHub\]](#) [\[Demo\]](#)
- Architected a distributed application security intelligence platform across 17 monorepo workspaces, achieving 100% precision on benchmark suites.
 - Built an AST JavaScript data-flow engine detecting DOM XSS without browser overhead; engineered multi-identity BOLA/IDOR verification matrix.
 - Constructed Attack-Surface Graph 2.0 with Cytoscape topology visualization, automated secret redaction, and SARIF v2.1.0 report generation.
- StackLens — Website Architecture Inference & Intelligence Engine | Java 21, Spring Boot 3, RabbitMQ, PostgreSQL, Redis, React 19, TypeScript** [\[GitHub\]](#) [\[Demo\]](#)
- Architected an asynchronous multi-vector probe engine reverse-engineering web architectures across 200+ signatures with RFC 1918 SSRF defense.
 - Decoupled external crawlers from APIs via RabbitMQ & Redis; synthesized PostgreSQL schemas and interactive DAG topologies.
- LiquidityLens — Market Microstructure Simulator & Matching Engine | C++17, Python 3.10, FastAPI, Fixed-Point Math, Hawkes Processes, WebSockets** [\[GitHub\]](#) [\[Demo\]](#)
- Developed a deterministic C++17 LOB matching engine supporting 4.5M+ events/s with ~220ns match latency using `int64_t` fixed-point arithmetic.
 - Simulated Hawkes process order clusters, modeled FIFO queue priority across 10μs–500μs latency slips, and calculated adverse selection markouts.
- TaskFlow — Real-Time Collaborative Kanban & Distributed State Engine | React 18, Node.js, PostgreSQL 17, Prisma ORM, Socket.io, TanStack Query** [\[GitHub\]](#) [\[Demo\]](#)
- Engineered a real-time collaborative workspace with Optimistic Concurrency Control (OCC) using integer revision tags to reject stale writes.
 - Implemented fractional midpoint indexing for O(1) card drag reordering and room-scoped Socket.io state synchronization.

EDUCATION

- Presidency University — Bengaluru, India** Sept 2023 – Present | Expected 2027
- Bachelor of Technology (B.Tech) in Computer Science & Engineering*
- Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems (DBMS), Computer Networks, System Design, Software Engineering.

PROBLEM SOLVING & CERTIFICATIONS

- Competitive Programming:** Active problem solver on [LeetCode \(u/playboldAbhi\)](https://leetcode.com/u/playboldAbhi) and [Codeforces \(playboldAbhi\)](https://codeforces.com/abhi), focusing on Graph Algorithms, DP, and Amortized Complexity.
- HackerRank Certifications:** SQL (Advanced), Rest API (Intermediate), Problem Solving (Intermediate), Java (Basic).